

Retail Product Sales Analysis Using Excel - Data Exploration and Insights

Project Objective:

To explore a retail product sales dataset using Excel functions - SUM, COUNT, AVERAGE, MIN, MAX, IF, SUMIF, COUNTIF, LEFT, RIGHT, and MID - and extract meaningful business insights from the data.

Dataset Description:

The dataset is a retail sales dataset containing 34 product records. It covers multiple product categories such as Electronics, Fashion, Kitchen, Outdoor, and Accessories. The products are sold across various countries including US, UK, Canada, India, Australia, Germany, Italy, France, Spain, China, Russia, and Brazil. Each record includes details like the product name, brand, price, quantity, and category.

Column Description

Column Name	Description
Product ID	Unique identifier combining date and country code (e.g. 28-JAN-US)
Product Name	Name of the product (e.g. Laptop, Sneakers, Blender)
Brand Name	Brand of the product (e.g. Dell, Nike, Ninja)
Price (\$)	Selling price of the product in US dollars
Quantity	Number of units sold
Category	Product category - Electronics, Fashion, Kitchen, Outdoor, Accessories
Price Range	High Price (≥ 500) or Standard Price (< 500) - derived using IF formula
Day	Day extracted from Product ID using LEFT formula
Country Code	Country code extracted from Product ID using RIGHT formula
Month	Month extracted from Product ID using MID formula

Data Cleaning and Transformation:

The dataset did not have any missing values. A new column called Price Range was added using the IF formula to classify products as High Price or Standard Price based on their price value. Three additional columns - Day, Country Code, and Month - were created by extracting parts of the Product ID using the LEFT, RIGHT, and MID text functions.

Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

Descriptive Analysis:

Total price of all 34 products is 10100. The average price is 297.06. The dataset has products across 5 categories and 12 countries.

Diagnostic Analysis:

The minimum price in the dataset is 30 (T-shirt by Adidas) and the maximum price is 1000 (Laptop by Dell). Electronics category alone contributes 8050 out of the total 10100 revenue.

Predictive Analysis:

Since Electronics generates close to 80% of total revenue, future stock and sales efforts should focus more on the Electronics category. Budget products priced under 100 have high volume potential.

Prescriptive Analysis:

Running discount campaigns on the 11 products priced below 100 can increase volume sold. Maintaining strong stock levels for high-priced Electronics like Laptops and Smartphones is recommended as they carry the highest revenue value.

Quick Reference - All Formulas

Function	Formula Used	Result
SUM	=SUM(D2:D35)	10100
COUNT	=COUNT(D2:D35)	34
AVERAGE	=AVERAGE(D2:D35)	297.06
MIN	=MIN(D2:D35)	30 - T-shirt by Adidas
MAX	=MAX(D2:D35)	1000 - Laptop by Dell
IF	=IF(D2>=500,"High Price","Standard Price")	High Price or Standard Price
SUMIF	=SUMIF(F2:F35,"Electronics",D2:D35)	8050
COUNTIF	=COUNTIF(D2:D35,"<100")	11
LEFT	=LEFT(A2,2)	28 - Day
RIGHT	=RIGHT(A2,2)	US - Country Code
MID	=MID(A2,4,3)	JAN - Month

Operators inside IF, SUMIF, COUNTIF:

Operator	What It Means
>	Greater than - example: >500 means more than 500
<	Less than - example: <100 means less than 100
>=	Greater than or equal - example: >=500 means 500 or more
<=	Less than or equal - example: <=100 means 100 or less
=	Equal to - example: =500 means exactly 500 only
<>	Not equal to - example: <>500 means anything except 500

Task 1 - SUM, COUNT and AVERAGE

What does each function do?

SUM - Adds up all the numbers in a column. Like adding total marks.

COUNT - Counts how many numbers exist. Like counting how many students.

AVERAGE - Finds the middle value. Excel adds everything and divides automatically.

Formula

=SUM(D2:D35) - adds all prices from row 2 to row 35

=COUNT(D2:D35) - counts how many products have a price

=AVERAGE(D2:D35) - finds the average price

Results

Formula	What It Does	Result
=SUM(D2:D35)	Total of all product prices	10100
=COUNT(D2:D35)	Number of products in the list	34
=AVERAGE(D2:D35)	Average price of all products	297.06

Fig 1 and Fig 2 - Task 1: SUM =SUM(D2:D35) | COUNT =COUNT(D2:D35)

Q1 - SUM: Calculate the TOTAL price of all products in the dataset.

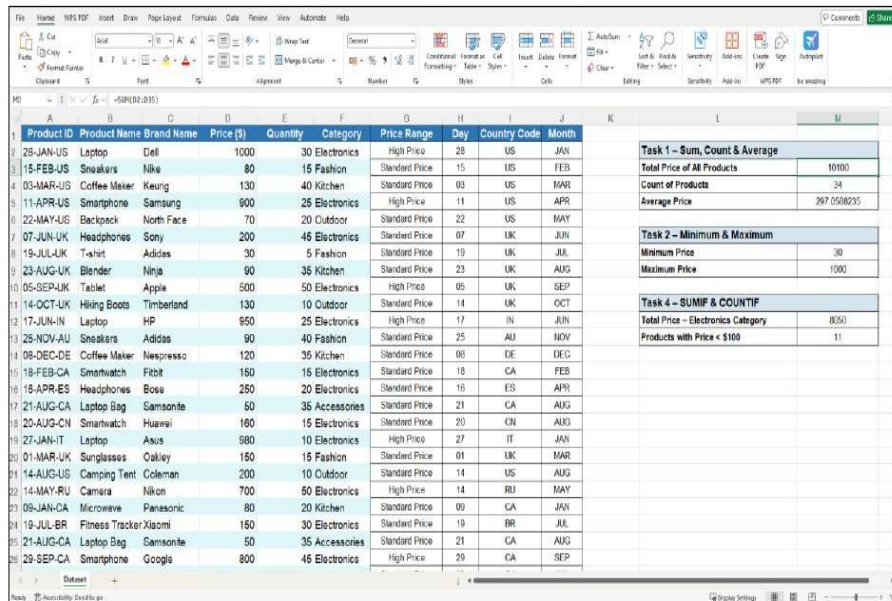
Formula: =SUM(D2:D35) | Result: Total of all product prices = 10100

Q2 - COUNT: Count HOW MANY products exist in the dataset.

Formula: =COUNT(D2:D35) | Result: Number of products = 34

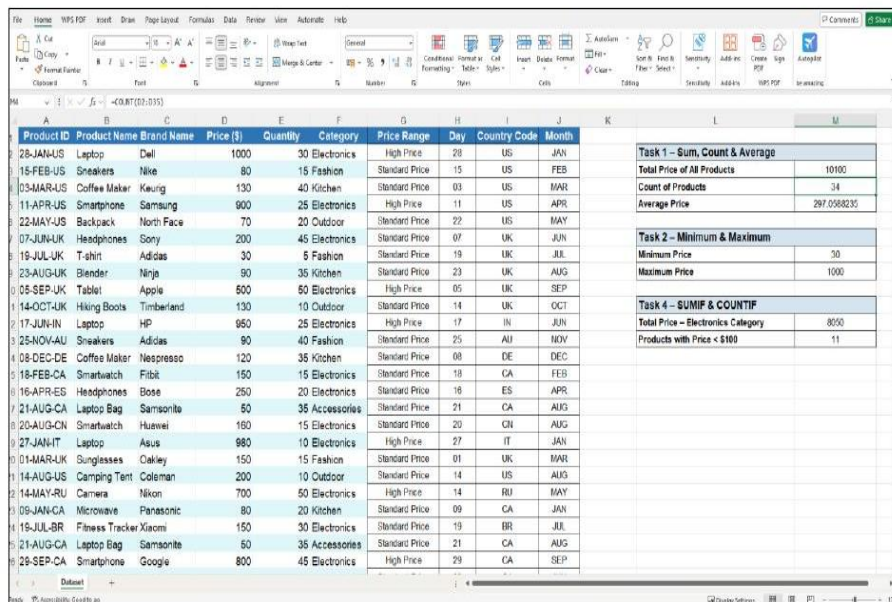
Data Analytics (DA) — Module 1: Excel Assignment 1 – Data Exploration | Formula

SCREENSHOTS TASK 1 — BASIC DATA EXPLORATION: SUM, COUNT, AVERAGE FIG 1: CELL M3 SELECTED — FORMULA BAR SHOWS =SUM(D2:D35)



Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	180	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 1: CELL M3 SELECTED — FORMULA BAR SHOWS =SUM(D2:D35) | RESULT: 10100



Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
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03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	180	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

FIG 2: CELL M4 SELECTED — FORMULA BAR SHOWS =COUNT(D2:D35) | RESULT: 34

Fig 3 and Fig 4 - Task 1: AVERAGE =AVERAGE(D2:D35) | Task 2: MIN =MIN(D2:D35)

Q3 - AVERAGE: Find the AVERAGE price of all products in the dataset.

Formula: =AVERAGE(D2:D35) | Result: Average price of all products = 297.06 (i.e. 10100 / 34)

Q4 - MIN: Find the CHEAPEST product price in the dataset.

Formula: =MIN(D2:D35) | Result: Minimum price = 30 (T-shirt by Adidas)

FIG 3: CELL M5 SELECTED — FORMULA BAR SHOWS =AVERAGE(D2:D35) | RESULT: 297.06

TASK 2 — MINIMUM & MAXIMUM VALUES: MIN, MAX

FIG 4: CELL M9 SELECTED — FORMULA BAR SHOWS =MIN(D2:D35) | RESULT: 30

Task 2 - MIN and MAX

What does each function do?

MIN - Finds the smallest number, finding the cheapest product.

MAX - Finds the largest number, finding the most expensive product.

Formula

=MIN(D2:D35) - finds the lowest price in the list

=MAX(D2:D35) - finds the highest price in the list

Results

Formula	What It Finds	Result
=MIN(D2:D35)	Cheapest product price	30 - T-shirt by Adidas
=MAX(D2:D35)	Most expensive product price	1000 - Laptop by Dell

Fig 5 and Fig 6 - Task 2: MAX =MAX(D2:D35) | Task 3: IF Formula Applied to G2:G35

Q5 - MAX: Find the MOST EXPENSIVE product price in the dataset.

Formula: =MAX(D2:D35) | Result: Maximum price = 1000 (Laptop by Dell)

Q6 - IF: Mark products priced >= 500 as High Price and the rest as Standard Price.

Formula: =IF(D2>=500,"High Price","Standard Price") | Result: Applied to G2:G35 - classifies all 34 products

The screenshot shows an Excel spreadsheet with a dataset of products and their prices. The formula bar at the top displays '=MAX(D2:D35)' and the result '1000' is shown in cell M10. The dataset includes columns for Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The Price Range column is currently blank.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Kaung	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	180	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	80	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	20	CA	SEP

FIG 5: CELL M10 SELECTED — FORMULA BAR SHOWS =MAX(D2:D35) | RESULT: 1000

Task 3 — Logical Function: IF (Price Range Column)

The screenshot shows the same Excel spreadsheet as Figure 5, but with the formula bar displaying '=IF(D2>=500,"High Price","Standard Price")'. The result 'High Price' is shown in cell G2. The Price Range column is now populated for all products.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Kaung	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	180	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	80	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	20	CA	SEP

FIG 6: CELL G2 SELECTED — FORMULA BAR SHOWS =IF(D2>=500,"High Price","Standard Price")
| APPLIED TO G2:G35

Task 4 — Conditional Functions: SUMIF & COUNTIF

Task 3 - IF Function

What does IF do?

IF asks a YES or NO question about a cell and gives a different answer for each.

Example question: Is the price 500 or more?

If YES - write High Price in the cell

If NO - write Standard Price in the cell

Formula

=IF(D2>=500,"High Price","Standard Price")

3 parts

Part	What It Is	Example
Part 1 - Condition	The YES or NO question	D2>=500 - Is D2 bigger than or equal to 500?
Part 2 - YES answer	What to write if YES is true	High Price - must be in quotes
Part 3 - NO answer	What to write if YES is false	Standard Price - must be in quotes

Results

Product	Price	Result in Column G
Laptop - Dell	1000	High Price - 1000 >= 500 = YES
Sneakers - Nike	80	Standard Price - 80 >= 500 = NO
Smartphone - Samsung	900	High Price - 900 >= 500 = YES
Backpack - North Face	70	Standard Price - 70 >= 500 = NO
Tablet - Apple	500	High Price - 500 >= 500 = YES

Task 4 - SUMIF and COUNTIF

What does each function do?

SUMIF - Adds numbers only for rows that match a condition you set.

COUNTIF - Counts cells only when they match a condition you set.

Formula

=SUMIF(F2:F35,"Electronics",D2:D35) - total price of Electronics only

=COUNTIF(D2:D35,"<100") - count products priced below 100

SUMIF has 3 parts

Part	What It Means	In Your Assignment
Part 1 - Range	WHERE to look for the condition	F2:F35 - Category column
Part 2 - Criteria	WHAT to look for	"Electronics" - the category name
Part 3 - Sum Range	WHAT to add up when found	D2:D35 - Price column

COUNTIF has 2 parts only

Part	What It Means
Part 1 - Range	D2:D35 - Price column - where to look
Part 2 - Criteria	"<100" - count only prices less than 100

Results

Formula	Question It Answers	Result
=SUMIF(F2:F35,"Electronics",D2:D35)	Total price of Electronics products only	8050
=COUNTIF(D2:D35,"<100")	How many products cost less than 100	11

Fig 7 and Fig 8 - Task 4: SUMIF =SUMIF(F2:F35,"Electronics",D2:D35) | COUNTIF =COUNTIF(D2:D35,"<100")

Q7 - SUMIF: What is the total price of ALL Electronics products in the dataset?

Formula: =SUMIF(F2:F35,"Electronics",D2:D35) | Result: Total price of Electronics category = 8050

Q8 - COUNTIF: How many products cost LESS than 100 in the dataset?

Formula: =COUNTIF(D2:D35,"<100") | Result: Count of products priced below 100 = 11

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	City	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	38	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
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07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
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21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Acer	980	10	Electronics	High Price	27	IT	JAN
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14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
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19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
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29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

Task 1 - Sum, Count & Average	
Total Price of All Products	10100
Count of Products	34
Average Price	297.0588235

Task 2 - Minimum & Maximum	
Minimum Price	30
Maximum Price	1000

Task 4 - SUMIF & COUNTIF	
Total Price - Electronics Category	8050
Products with Price < \$100	11

FIG 7: CELL M13 SELECTED — FORMULA BAR SHOWS =SUMIF(F2:F35,"ELECTRONICS",D2:D35) | RESULT: 8050

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	City	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	38	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
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19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
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27-JAN-IT	Laptop	Acer	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP

Task 1 - Sum, Count & Average	
Total Price of All Products	10100
Count of Products	34
Average Price	297.0588235

Task 2 - Minimum & Maximum	
Minimum Price	30
Maximum Price	1000

Task 4 - SUMIF & COUNTIF	
Total Price - Electronics Category	8050
Products with Price < \$100	11

FIG 8: CELL M14 SELECTED — FORMULA BAR SHOWS =COUNTIF(D2: D35,"<100") | RESULT: 11

Task 5 — Text Functions: LEFT, RIGHT, MID

Task 5 - LEFT, RIGHT and MID

What does each function do?

- Your Product ID looks like this: 28-JAN-US
- It contains 3 pieces of information hidden inside it:
 - 28** - Day - first 2 characters from the LEFT
 - JAN** - Month - 3 characters from the MIDDLE starting at position 4
 - US** - Country Code - last 2 characters from the RIGHT

Formula

=LEFT(A2,2) - takes first 2 characters - Day (28)

=RIGHT(A2,2) - takes last 2 characters - Country Code (US)

=MID(A2,4,3) - starts at position 4, takes 3 characters - Month (JAN)

3 formulas

Function	Formula	Column - Applied To
LEFT	=LEFT(A2,2)	Column H - Day - Applied to H2:H35
RIGHT	=RIGHT(A2,2)	Column I - Country Code - Applied to I2:I35
MID	=MID(A2,4,3)	Column J - Month - Applied to J2:J35

Results

Product ID	Formula Used	Result
28-JAN-US	=LEFT(A2,2)	28 - Day
28-JAN-US	=MID(A2,4,3)	JAN - Month
28-JAN-US	=RIGHT(A2,2)	US - Country Code
15-FEB-US	=LEFT(A2,2)	15
07-JUN-UK	=RIGHT(A2,2)	UK
14-MAY-RU	=MID(A2,4,3)	MAY

Fig 9 and Fig 10 - Task 5: LEFT =LEFT(A2,2) Extracts Day | RIGHT =RIGHT(A2,2) Extracts Country Code

Q9 - LEFT: Extract the Day from the Product ID column (first 2 characters).

Formula: =LEFT(A2,2) | Result: Extracts Day from Product ID (e.g. 28-JAN-US = 28) - Applied to H2:H35

Q10 - RIGHT: Extract the Country Code from the Product ID column (last 2 characters).

Formula: =RIGHT(A2,2) | Result: Extracts Country Code from Product ID (e.g. 28-JAN-US = US) - Applied to I2:I35

The screenshot shows the Excel interface with the formula bar displaying '=LEFT(A2,2)'. The spreadsheet contains a table with columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The 'Day' column (H) contains the first two characters of the Product ID. The 'Country Code' column (I) contains the last two characters of the Product ID. The 'Month' column (J) contains the last three characters of the Product ID. The formula bar is set to '=LEFT(A2,2)'.

FIG 9: CELL H2 SELECTED — FORMULA BAR SHOWS =LEFT(A2,2) | EXTRACTS DAY (FIRST 2 CHARS) | APPLIED TO H2:H35

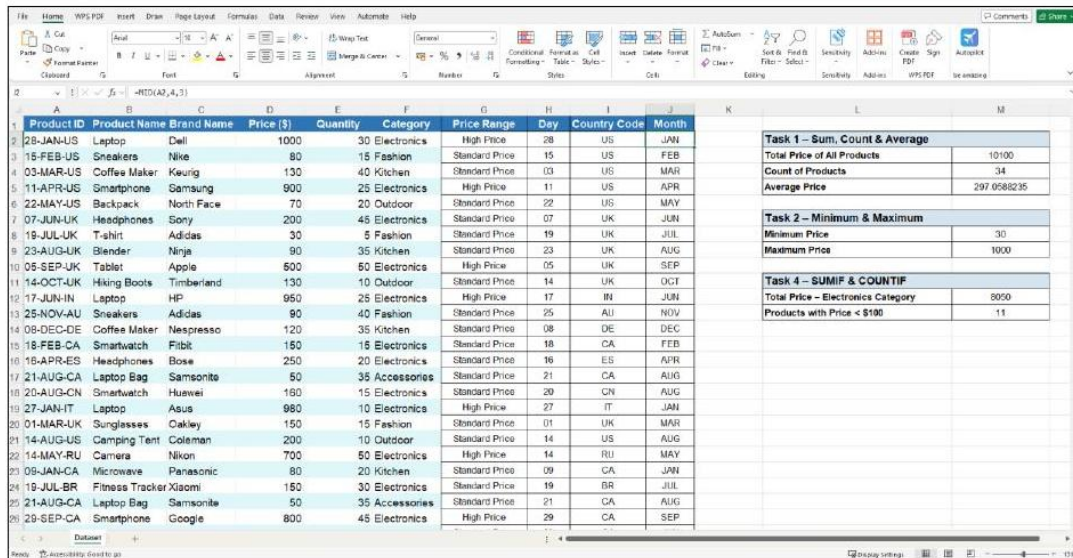
The screenshot shows the Excel interface with the formula bar displaying '=RIGHT(A2,2)'. The spreadsheet contains the same table as Figure 9. The 'Country Code' column (I) contains the last two characters of the Product ID. The 'Month' column (J) contains the last three characters of the Product ID. The formula bar is set to '=RIGHT(A2,2)'.

FIG 10: CELL I2 SELECTED — FORMULA BAR SHOWS =RIGHT(A2,2) | EXTRACTS COUNTRY CODE (LAST 2 CHARS) | APPLIED TO I2:I35

Fig 11 - Task 5: MID =MID(A2,4,3) Extracts Month from Product ID | Applied to J2:J35

Q11 - MID: Extract the Month from the Product ID column (start at position 4, take 3 characters).

Formula: =MID(A2,4,3) | Result: Extracts Month from Product ID (e.g. 28-JAN-US = JAN) - Applied to J2:J35



Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
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Task 2 – Minimum & Maximum	
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Task 4 – SUMIF & COUNTIF	
Total Price – Electronics Category	8050
Products with Price < \$190	11

FIG 11: CELL J2 SELECTED — FORMULA BAR SHOWS =MID(A2,4,3) | EXTRACTS MONTH (CHARS 4-6) |
APPLIED TO J2:J35

Conclusion:

This assignment helped in understanding how to explore and analyze a dataset using Excel formulas. All five tasks were completed successfully using SUM, COUNT, AVERAGE, MIN, MAX, IF, SUMIF, COUNTIF, LEFT, RIGHT, and MID functions. The analysis shows that Electronics is the top revenue-generating category at 8050 out of 10100 total. The dataset spans 12 countries and 5 product categories. Budget products under 100 make up 11 out of 34 items and can be targeted for volume-based sales strategies.